DECISION TREES

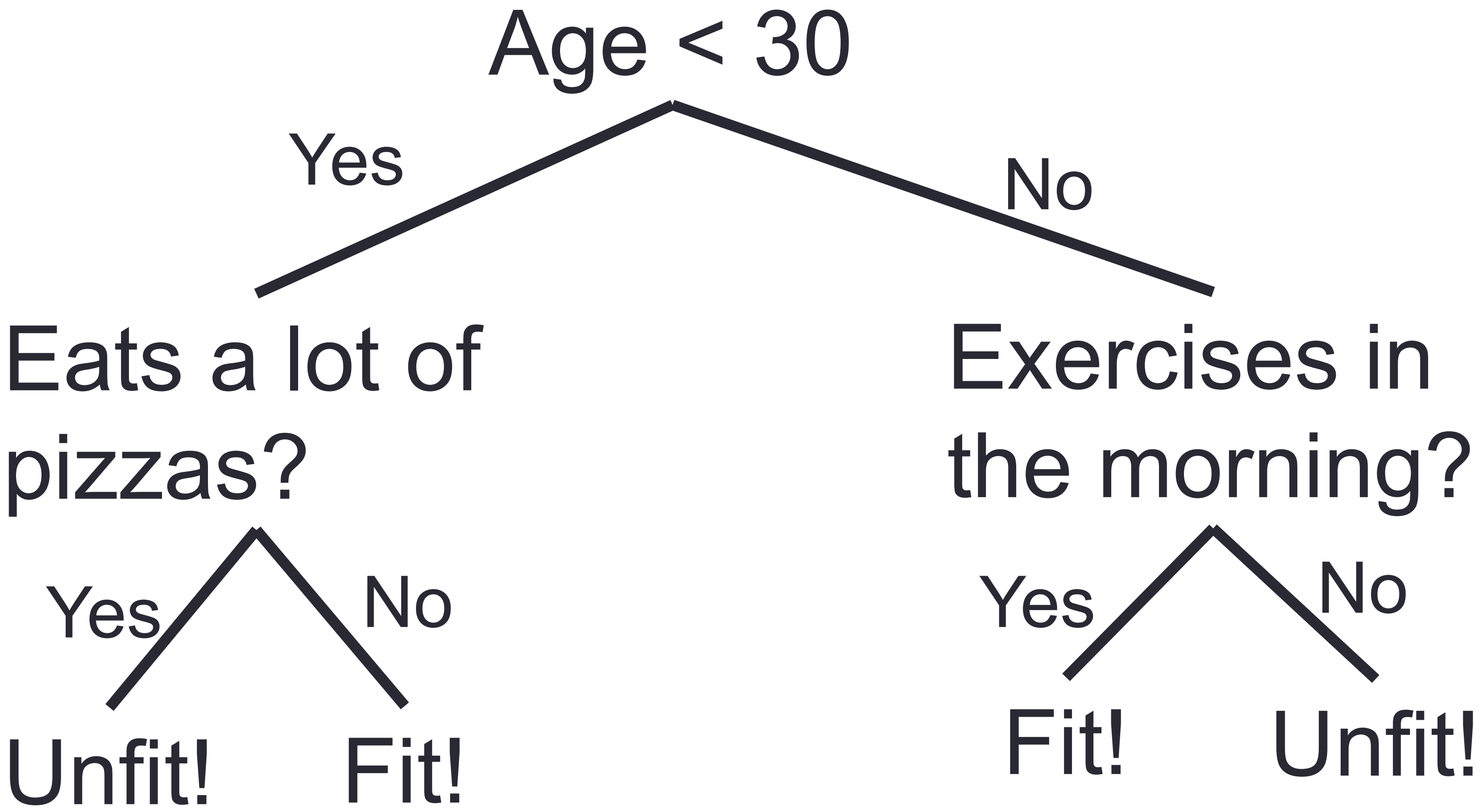
Decision Trees

- It is used for classification
- It is a type of Supervised Machine Learning
  - what the input is and what the corresponding output is in the training data.
  - where the data is continuously split according to a certain parameter.
- The tree can be explained by two entities, namely **decision nodes** and **leaves**.
- The leaves are the decisions or the final outcomes
- The decision nodes are where the data is split.

Example

- Assume that you want to predict whether a person is fit given their information like age, eating habit, and physical activity, etc.
- The decision nodes here are questions like ‘What’s the age?’, ‘Does he exercise?’, ‘Does he eat a lot of pizzas’?
- And the leaves, which are outcomes like either ‘fit’, or ‘unfit’. In this case this was a binary classification problem.

Example: Is a Person Fit?



Why decision trees?

- Decision trees often mimic the human level thinking so its so simple to understand the data and make some good interpretations.

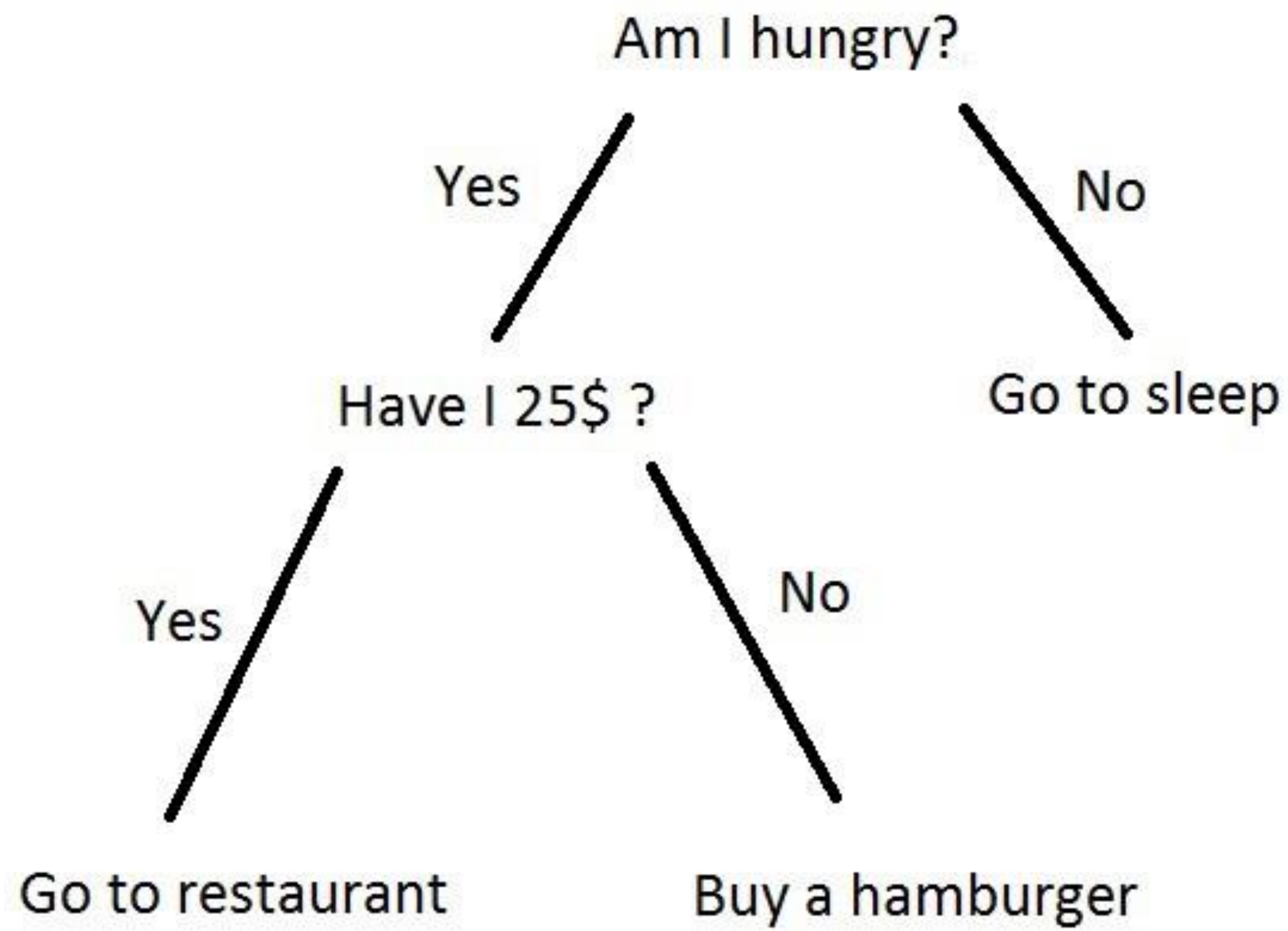
- Decision trees actually make you see the logic for the data to interpret (not like black box algorithms like SVM, NN, etc.)

Types of Decision Trees

- **Classification trees** (Yes/No types)
  - Previous example is an example of classification tree, where the outcome was a variable like ‘fit’ or ‘unfit’.
  - Here the decision variable is **Categorical**.

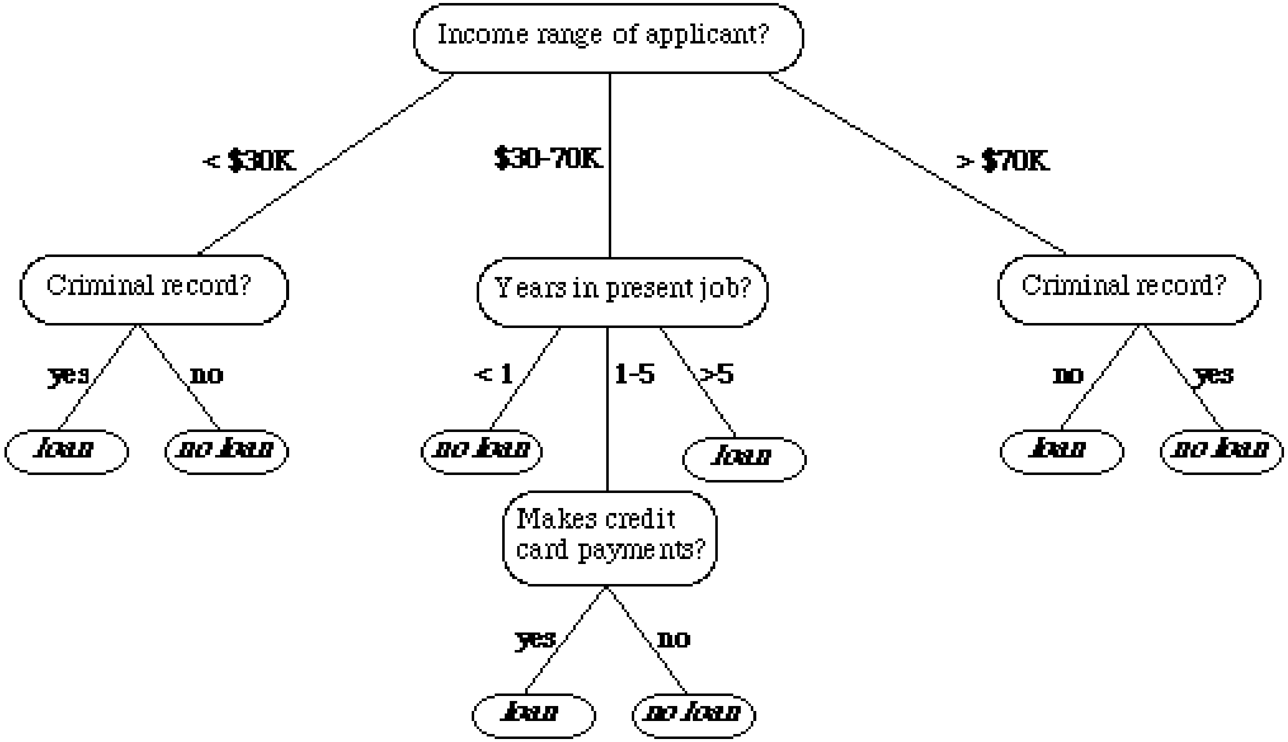
- **Regression trees** (Continuous data types)
  - Here the decision or the outcome variable is **Continuous**, e.g. a number like 123.

Few more Examples



Few more Examples

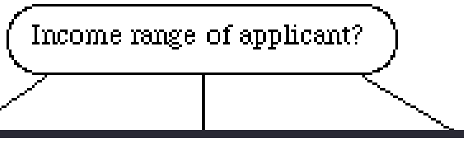
- If we are classifying *bank loan* application for a customer; the decision tree may look like this.



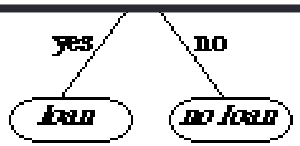
- Here we can see the logic how it is making the decision.

Few more Examples

- if we are classifying *bank loan* application for a customer, the decision tree may look like this.



A decision tree is a tree where each node represents a feature(attribute), each link(branch) represents a decision(rule) and each leaf represents an outcome(categorical or continues value).



- Here we can see the logic how it is making the decision.

Working

- There are many algorithms out there which construct Decision Trees, but one of the best is called as **ID3 Algorithm**.

- ID3 Stands for **Iterative Dichotomiser 3**.

- Before discussing the ID3 algorithm, we’ll go through few definitions.

Entropy

- Entropy, also called as Shannon Entropy
- It is the measure of the amount of uncertainty or randomness in data.

Entropy

- Example, consider a coin toss whose probability of heads is 0.5 and probability of tails is 0.5.

- Here the entropy is the highest possible, since there's no way of determining what the outcome might be.

- Alternatively, consider a coin which has heads on both the sides, the entropy of such an event can be predicted perfectly since we know beforehand that it'll always be heads. In other words, this event has **no randomness** hence its entropy is zero.

- In particular, lower values imply less uncertainty while higher values imply high uncertainty.

Information Gain

- Information gain is also called as Kullback-Leibler divergence

- It measures the relative change in entropy with respect to the independent variables.

Example

- Consider a piece of data collected over the course of 14 days where the features are Outlook, Temperature, Humidity, Wind and the outcome variable is whether Golf was played on the day.
- Now, our job is to build a predictive model which takes in above 4 parameters and predicts whether Golf will be played on the day.

- We will build a decision tree to do that using **ID3 algorithm.**

Example

Day	Outlook	Temperature	Humidity	Wind	Play Golf
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

ID3 Algorithm

1. Create root node for the tree

   The attribute that best classifies the training data, use this attribute at the root of the tree

- How to choose the best attribute?
  - ID3 algorithm begins

ID3 Algorithm

- Calculate the entropy (Amount of uncertainty in dataset)

$$Entropy = \frac{-p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

- Calculate Average Information

$$I(Attribute) = \sum \frac{p_i + n_i}{p+n} Entropy(A)$$

- Calculate Information Gain (Difference in entropy before and after splitting dataset on attribute A)

$$Gain = Entropy(S) - I(Attribute)$$

ID3 Algorithm

1. Compute the Entropy for dataset entropy(S)
2. For every attribute/ feature:
   1. Calculate entropy for all other values Entropy (A)
   2. Take average information entropy for the current attribute
   3. Calculate Gain for the current attribute
3. Pick the highest Gain attribute
4. Repeat until we get the tree we desired

Example

Day	Outlook	Temperat ure	Humidity	Wind	Play Golf
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

P = 9

N = 5

Total = 14

Example

P	N	Total
9	5	14

$$Entropy = \frac{-p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

$$Entropy = \frac{-9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$= 0.940$$

Example - For Each Attribute (let say outlook)

Calculate Entropy for each Values i.e. for Sunny, Rain, , Overcast

Outlook	Play Golf
Sunny	No
Sunny	No
Sunny	No
Sunny	Yes
Sunny	Yes

Outlook	Play Golf
Rain	Yes
Rain	Yes
Rain	No
Rain	Yes
Rain	No

Outlook	Play Golf
Overcast	Yes
Overcast	Yes
Overcast	Yes
Overcast	Yes

Outlook	P	N	Entropy
Sunny	2	3	0.971
Rain	3	2	0.971
Overcast	4	0	0

Calculate Entropy (Outlook = 'Value')

$$Entropy = \frac{-p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

$$Entropy(Outlook = sunny) = \frac{-2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right) = 0.971$$

$$Entropy(Outlook = overcast) = 1\log_2(1) - 0\log_2 0 = 0$$

$$Entropy(Outlook = rain) = \frac{-3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right) = 0.971$$

Calculate Average Information (Entropy)

$$I(Outlook) = \frac{P_{sunny} + N_{sunny}}{p+n} Entropy(Outlook = Sunny) +$$

$$\frac{P_{rainy} + N_{rainy}}{p+n} Entropy(Outlook = rainy) +$$

$$\frac{P_{outcast} + N_{outcast}}{p+n} Entropy(Outlook = outcast)$$

$$I(Outlook) = \frac{3+2}{9+5} * 0.971 + \frac{2+3}{9+5} * 0.971 + \frac{4+0}{9+5} * 0$$

$$= 0.693$$

Calculate Gain: attribute is outlook

$$Gain = Entropy(S) - I(Attribute)$$

$$Entropy(S) = 0.940$$

$$Gain(Outlook) = 0.940 - 0.693 = 0.247$$

Attribute (Temperature)

Calculate Entropy for each Temperature i.e. for Hot, Mild, Cool

Temperature	Play Golf
Hot	No
Hot	No
Hot	Yes
Hot	Yes

Temperature	Play Golf
Mild	Yes
Mild	No
Mild	Yes
Mild	Yes
Mild	Yes
Mild	No

Temperature	Play Golf
Cool	Yes
Cool	No
Cool	Yes
Cool	Yes

Temperature	P	N	Entropy
Hot	2	2	1
Mild	4	2	0.918
Cool	3	1	0.811

Calculate Average Information (Entropy)

$$I(Temperature) = \frac{P_{hot} + N_{hot}}{p+n} Entropy(Temperature = hot) +$$

$$\frac{P_{mild} + N_{mild}}{p+n} Entropy(Temperature = mild) +$$

$$\frac{P_{cool} + N_{cool}}{p+n} Entropy(Temperature = cool)$$

$$I(Temperature) = \frac{2+2}{9+5} * 1 + \frac{4+2}{9+5} * 0.918 + \frac{3+1}{9+5} * 0.811$$

$$= 0.911$$

Calculate Gain: attribute is outlook

$$Gain = Entropy(S) - I(Attribute)$$

$$Entropy(S) = 0.940$$

$$Gain(Temperature) = 0.940 \; - \; 0.911 \; = \; 0.029$$

Attribute (humidity)

Humidity	Play Golf
High	No
High	No
High	Yes
High	Yes
High	No
High	Yes
High	No

Humidity	Play Golf
Normal	Yes
Normal	No
Normal	Yes
Normal	Yes
Normal	Yes
Normal	Yes
Normal	Yes

Humidity	P	N	Entropy
High	3	4	0.985
Normal	6	1	0.591

Calculate Average Information (Entropy)

$$I(Humidity) = \frac{P_{high} + N_{high}}{p+n} Entropy(Humidity = high) +$$

$$\frac{P_{normal} + N_{normal}}{p+n} Entropy(Humidity = normal)$$

$$I(Humidity) = \frac{3+4}{9+5} * 0.985 + \frac{6+1}{9+5} * 0.591$$

$$= 0.788$$

Calculate Gain: attribute is outlook

$$Gain = Entropy(S) - I(Attribute)$$

$$Entropy(S) = 0.940$$

$$Gain(Humidity) = 0.940 \ - \ 0.788 \ = \ 0.152$$

Attribute (Wind)

Wind	Play Golf
Weak	No
Weak	Yes
Weak	Yes
Weak	Yes
Weak	No
Weak	Yes
Weak	Yes
Weak	Yes

Wind	Play Golf
Strong	No
Strong	No
Strong	Yes
Strong	Yes
Strong	Yes
Strong	No

Wind	P	N	Entropy
Weak	6	2	0.811
Strong	3	3	1

Calculate Average Information (Entropy)

$$I(Wind) = \frac{P_{strong} + N_{strong}}{p+n} Entropy(Wind = strong) +$$

$$\frac{P_{weak} + N_{weak}}{p+n} Entropy(Wind = weak)$$

$$I(Wind) = \frac{3+3}{9+5} * 1 + \frac{6+2}{9+5} * 0.811$$

$$= 0.892$$

Calculate Gain: attribute is outlook

$$Gain = Entropy(S) - I(Attribute)$$

$$Entropy(S) = 0.940$$

$$Gain(wind) = 0.940 \ - \ 0.892 \ = \ 0.048$$

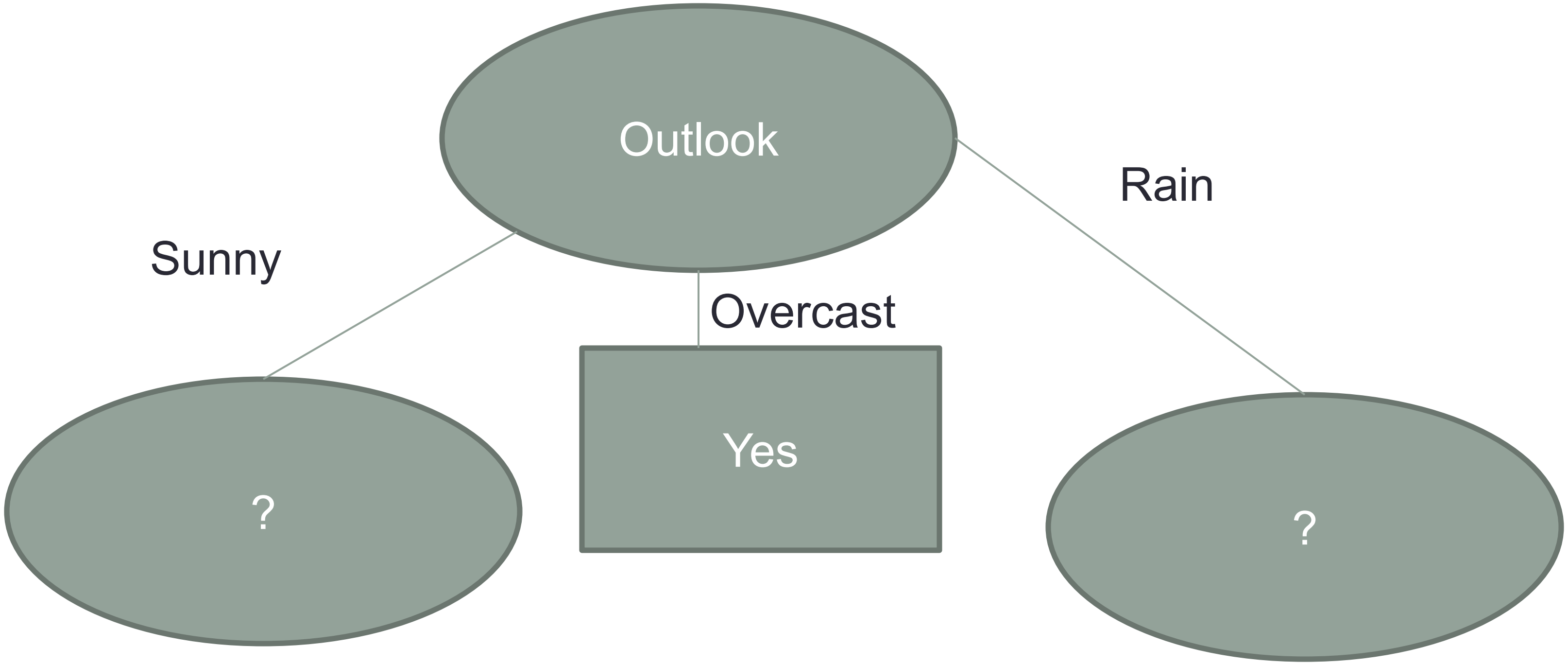
Pick the highest Gain Attribute

Attribute	Gain
Outlook	0.247
Temperature	0.029
Humidity	0.152
Wind	0.048

Root Node

OUTLOOK

Decision Tree



Repeat the same thing for sub trees till we get the tree

Outlook	Temperature	Humidity	Wind	Play Golf
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

- Outlook = Sunny

Outlook	Temperature	Humidity	Wind	Play Golf
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	Strong	No

- Outlook = rain

Repeat the same thing for sub trees till we get the tree

Outlook	Temperature	Humidity	Wind	Play Golf
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

- Outlook = Sunny
- P=2 N=3
- Total = 5

$$Entropy = \frac{-p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

$$Entropy = \frac{-2}{2+3}\log_2\left(\frac{2}{2+3}\right) - \frac{3}{2+3}\log_2\left(\frac{3}{2+3}\right)$$

$$= 0.971$$

Repeat the same thing for sub trees till we get the tree

Outlook	Temperature	Play Golf
Sunny	Hot	No
Sunny	Hot	No
Sunny	Mild	No
Sunny	Cool	Yes
Sunny	Mild	Yes

Temperature	P	N	Entropy
Cool	1	0	0
Hot	0	2	0
Mild	1	1	1

Calculate Average Information Entropy

$$I(Temperature) = 0.4$$

Calculate Gain

$$Gain = 0.571$$

Repeat the same thing for sub trees till we get the tree

Outlook	Humidity	Play Golf
Sunny	High	No
Sunny	High	No
Sunny	High	No
Sunny	Normal	Yes
Sunny	Normal	Yes

Humidity	P	N	Entropy
High	0	3	0
Normal	2	0	0

Calculate Average Information Entropy

$$I(Humidity) = 0$$

Calculate Gain

$$Gain = 0.971$$

Repeat the same thing for sub trees till we get the tree

Outlook	Wind	Play Golf
Sunny	Weak	No
Sunny	Strong	No
Sunny	Weak	No
Sunny	Weak	Yes
Sunny	Strong	Yes

Wind	P	N	Entropy
Stron g	1	1	1
Weak	1	2	0.918

Calculate Average Information Entropy

$$I(Wind) = 0.951$$

Calculate Gain

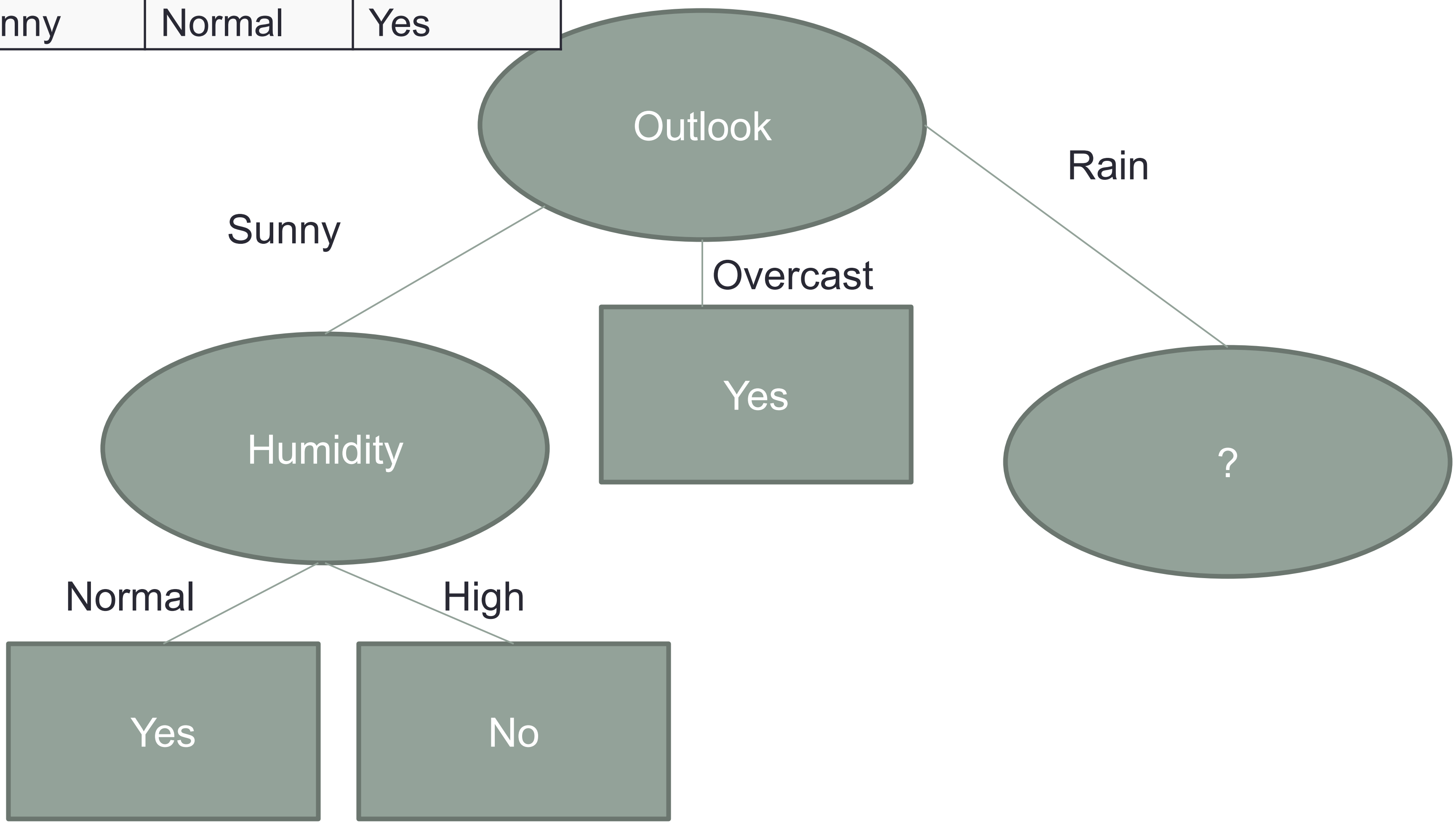
$$Gain = 0.020$$

Pick the highest Gain Attribute

Attribute	Gain
Temperature	0.571
Humidity	0.971
Wind	0.02

Root Node
Humidity

Outlook	Humidity	Play Golf
Sunny	High	No
Sunny	High	No
Sunny	High	No
Sunny	Normal	Yes
Sunny	Normal	Yes



Outlook	Temperature	Humidity	Wind	Play Golf
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	Strong	No

- Outlook = Rain
- P=3 N=2
- Total = 5

$$Entropy = \frac{-p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

$$Entropy = \frac{-3}{2+3}\log_2\left(\frac{3}{2+3}\right) - \frac{2}{2+3}\log_2\left(\frac{2}{2+3}\right)$$

$$= 0.971$$

For Each Attribute (Temperature)

Outlook	Temperature	Play Golf
Rain	Mild	Yes
Rain	Cool	Yes
Rain	Cool	No
Rain	Mild	Yes
Rain	Mild	No

Temperature	P	N	Entropy
Mild	2	1	0.918
Cool	1	1	1

Calculate Average Information Entropy

$$I(Temperature) = 0.951$$

Calculate Gain

$$Gain = 0.020$$

For Each Attribute (Humidity)

Outlook	Humidity	Play Golf
Rain	High	Yes
Rain	Normal	Yes
Rain	Normal	No
Rain	Normal	Yes
Rain	High	No

Humidity	P	N	Entropy
High	1	1	1
Normal	2	1	0.918

Calculate **Average Information Entropy**

$$I(Humidity) = 0.951$$

Calculate **Gain**

$$Gain = 0.020$$

For Each Attribute (Wind)

Outlook	Wind	Play Golf
Rain	Weak	Yes
Rain	Weak	Yes
Rain	Strong	No
Rain	Weak	Yes
Rain	Strong	No

Wind	P	N	Entropy
Strong	0	2	0
Weak	3	0	0

Calculate Average Information Entropy

$$I(Wind) = 0$$

Calculate Gain

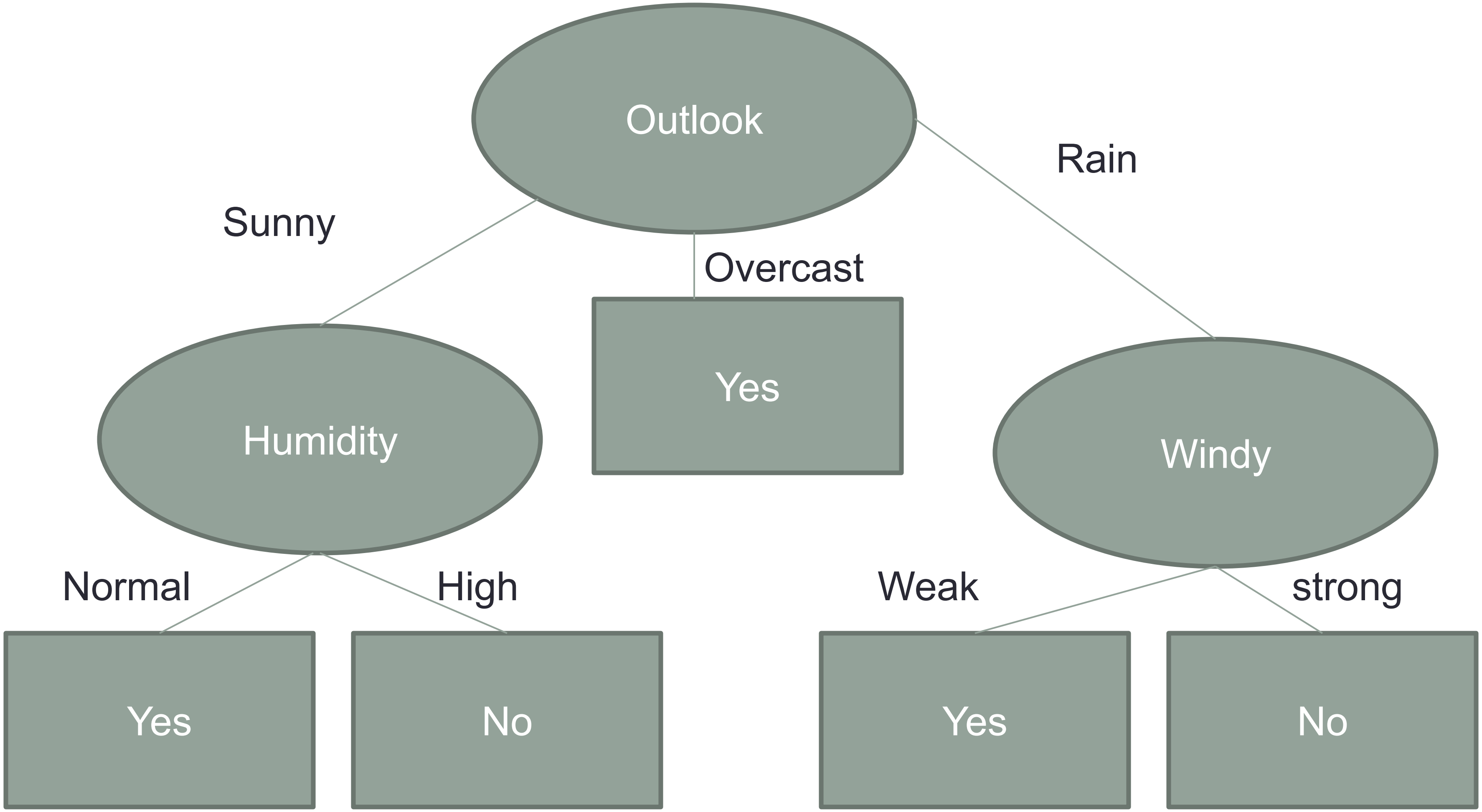
$$Gain = 0.971$$

Pick the highest Gain Attribute

Attribute	Gain
Temperature	0.02
Humidity	0.02
Wind	0.971

Root Node in Rainy
wind

Final Decision Tree



Practice Question

Apply Decision tree algorithm on following dataset:

Color	Shape	Size	Class
Red	Square	Large	Like
Blue	Square	Large	Like
Red	Round	Small	Dislike
Green	Square	Small	Dislike
Red	Round	Large	Like
Green	Round	Large	Dislike

Learning decision trees

Example Problem: Decide whether to wait for a table at a restaurant, based on the following attributes:

1. **Alternate**: is there an alternative restaurant nearby?
2. **Waiting area**: is there a comfortable area to wait in?
3. **Fri/Sat**: is today Friday or Saturday?
4. **Hungry**: are we hungry?
5. **Patrons**: number of people in the restaurant (None, Some, Full)
6. **Price**: price range ($, $$, $$$)
7. **Raining**: is it raining outside?
8. **Reservation**: have we made a reservation?
9. **Type**: kind of restaurant (French, Italian, Thai, Burger)
10. **Wait Estimate**: estimated waiting time (0-10, 10-30, 30-60, >60)

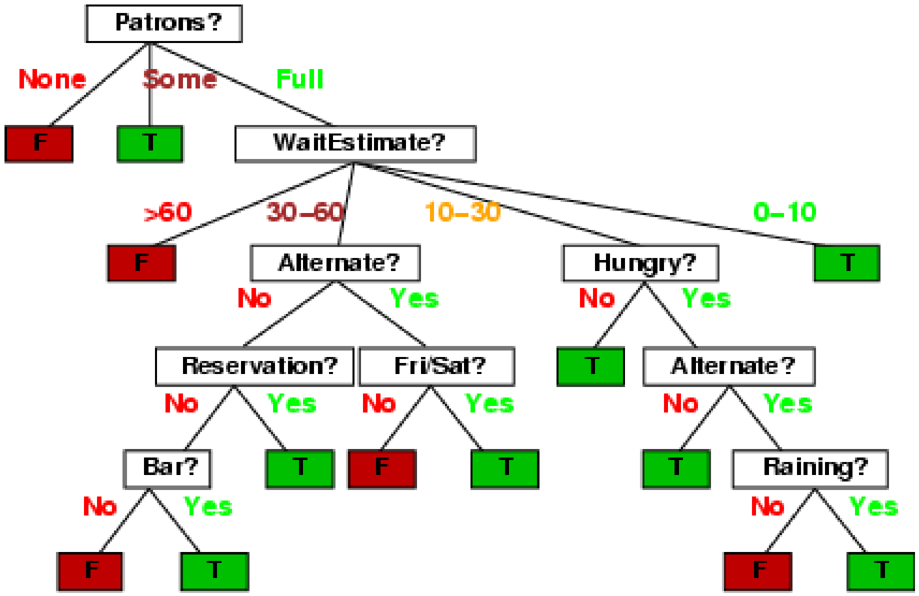
Feature(Attribute)-based representations

- Examples described by feature(attribute) values
  - (Boolean, discrete, continuous)
- E.g., situations where I will/won't wait for a table:

Example	Attributes										Target
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*Wait*
X_1	T	F	F	T	Some	$$$	F	T	French	0–10	T
X_2	T	F	F	T	Full	$	F	F	Thai	30–60	F
X_3	F	T	F	F	Some	$	F	F	Burger	0–10	T
X_4	T	F	T	T	Full	$	F	F	Thai	10–30	T
X_5	T	F	T	F	Full	$$$	F	T	French	>60	F
X_6	F	T	F	T	Some	$$	T	T	Italian	0–10	T
X_7	F	T	F	F	None	$	T	F	Burger	0–10	F
X_8	F	F	F	T	Some	$$	T	T	Thai	0–10	T
X_9	F	T	T	F	Full	$	T	F	Burger	>60	F
X_{10}	T	T	T	T	Full	$$$	F	T	Italian	10–30	F
X_{11}	F	F	F	F	None	$	F	F	Thai	0–10	F
X_{12}	T	T	T	T	Full	$	F	F	Burger	30–60	T

Decision trees

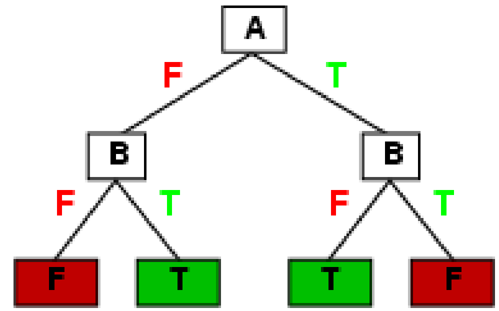
- One possible representation for hypotheses
- E.g., here is the “true” tree for deciding whether to wait:



Expressiveness

- Decision trees can express any function of the input attributes.
- E.g., for Boolean functions, truth table row → path to leaf:

A	B	A xor B
F	F	F
F	T	T
T	F	T
T	T	F



- Trivially, there is a consistent decision tree for any training set with one path to leaf for each example (unless *f* nondeterministic in *x*) but it probably won't generalize to new examples

- Prefer to find more compact decision trees

Decision Tree Based Classification

Advantages:

- **Easy** to construct/implement
- Extremely **fast** at classifying unknown records
- Models are **easy to interpret for small-sized trees**
- **Accuracy is comparable** to other classification techniques for many simple data sets
- Tree models make no assumptions about the distribution of the underlying data : **nonparametric**
- Have a **built-in feature selection** method that makes them immune to the presence of useless variables

Decision Tree Based Classification

Disadvantages:

- Computationally **expensive to train**
- **Some decision trees can be overly complex** that do not generalize the data well.
- **Less expressivity**: There may be concepts that are hard to learn with limited decision trees

Conclusion

- Below is the summary of what we’ve studied:
- Entropy to measure discriminatory power of an attribute for classification task. It defines the amount of randomness in attribute for classification task. Entropy is minimal means the attribute appears close to one class and have a good discriminatory power for classification

- Information Gain to rank attribute for filtering at given node in the tree. The ranking is based on high information gain entropy in decreasing order.

- The recursive ID3 algorithm that creates a decision tree.